Practice workbook · Forming a new quadratic from the roots of another

Building a quadratic equation, usually with integer coefficients, whose roots are given expressions in alpha and beta, using only the sum and product of the original roots.

22 question parts: 1 worked-example retry and 21 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-QUA-10

1. Question 6 · part c · 4 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

6 (a) Show that $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

(3)

The quadratic equation $x^2 - 7kx + k^2 = 0$, where k is a positive constant, has roots α and β where $\alpha > \beta$

(b) Show that $\alpha - \beta = 3k\sqrt{5}$

(3)

(c) Hence form a quadratic equation with roots $\alpha + 1$ and $\beta - 1$

Give your equation in the form $x^2 + px + q = 0$ where p and q should be given in terms of k.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 6 · part c · 4 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 6 · part c · 4 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 6 · part c · 4 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 6 is 10 marks)

1. Question 6 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 10 · part b · 7 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

10 The roots of a quadratic equation are α and β where

$$\alpha+\beta=-\frac{5}{2} \text{ and } \alpha^3+\beta^3=\frac{115}{8}$$

(a) Show that $\alpha\beta=4$

(3)

(b) Form a quadratic equation with integer coefficients, that has roots

$$\frac{\alpha^2+1}{\beta} \text{ and } \frac{\beta^2+1}{\alpha}$$

(7)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 3 5 8 6 A 0 2 6 3 2

2. Question 10 · part b · 7 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 10 · part b · 7 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 10 · part b · 7 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 10 is 10 marks)

2. Question 10 · part b · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 5 · part b · 6 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

5 (a) Show that $(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)=\alpha^3+\beta^3$ (2)

The quadratic equation $2x^2-6x-7=0$ has roots α and β

Without solving the equation

(b) form a quadratic equation, with integer coefficients, which has roots $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$ (6)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

P 7 2 8 6 5 A 0 1 6 3 6

3. Question 5 · part b · 6 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 5 is 8 marks)

3. Question 5 · part b · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 11 · part c · 5 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

11 The quadratic equation $x^2 - px + q = 0$ where $p > 0$, has roots α and β.

Given that $2\alpha\beta = 3$ and that $4(\alpha^2 + \beta^2) = k^2 - 6k - 3$ where $k > 3$

(a) (i) write down the value of q,

(ii) find an expression, in terms of k, for p.

(5)

Given also that $7\alpha\beta = 3(\alpha + \beta)$

(b) find the value of k.

(2)

(c) Hence form an equation, with integer coefficients, which has roots

$$\frac{\alpha}{\alpha + \beta} \text{ and } \frac{\beta}{\alpha + \beta}$$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

P 6 1 8 8 3 A 0 3 4 3 6

4. Question 11 · part c · 5 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 11 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 11 · part c · 5 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 11 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 11 is 12 marks)

4. Question 11 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 7 · part b · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

7 (a) Find the set of values of k for which the equation $kx^2 - 4x + 2k = 7$ has real roots

(4)

Given that the roots of the equation $kx^2 - 4x + 2k = 7$ are α and β,

(b) form a quadratic equation with roots $\frac{\alpha + 1}{\alpha}$ and $\frac{\beta + 1}{\beta}$

Give each coefficient in terms of k.

(8)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 7 · part b · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 7 · part b · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 7 · part b · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 7 is 12 marks)

5. Question 7 · part b · 8 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 4 · part c · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

4 The quadratic equation

$$2x^2 + 4x + 3 = 0$$

has roots α and β

(a) Without solving the equation, show that $\alpha^2 + \beta^2 = 1$

(4)

(b) Without solving the equation, find the value of $\alpha^4 + \beta^4$

(3)

(c) Hence form a quadratic equation with integer coefficients that has roots α^4 and β^4

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

6. Question 4 · part c · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

6. Question 4 · part c · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10

P 6 6 3 0 7 A 0 1 0 3 2

6. Question 4 · part c · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 4 is 10 marks)

6. Question 4 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 8 · part c · 6 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

8 The quadratic equation $3x^2 - kx - 1 = 0$, where k is a positive integer, has roots α and β

(a) Show that $\alpha^2 + \beta^2 = \dfrac{k^2 + 6}{9}$

(3)

Given that $\alpha^4 + \beta^4 = \dfrac{466}{81}$

(b) find the value of k

(5)

(c) Hence form an equation, with integer coefficients, which has roots

$$\frac{\alpha^3 + \beta}{\beta} \text{ and } \frac{\beta^3 + \alpha}{\alpha}$$

(6)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

7. Question 8 · part c · 6 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 8 · part c · 6 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 8 · part c · 6 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 8 is 14 marks)

7. Question 8 · part c · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 10 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

10 The quadratic equation $2x^2 + kx + 4 = 0$ has roots α and β such that

$$k < 0 \text{ and } \alpha > \beta$$

Given that $\alpha^2 - \beta^2 = \frac{7\sqrt{17}}{4}$

(a) show that $k = -7$

(8)

(b) Hence form a quadratic equation that has roots

$$(\alpha - \beta) \text{ and } (\alpha + \beta)$$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

8. Question 10 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 10 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 10 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 10 continued

OLevels.com SCHOOL

(Total for Question 10 is 12 marks)

8. Question 10 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 6 · part a · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

6 $f(x) = 4x^2 - 3x - 5$

The equation $f(x) = 0$ has roots α and β

Without solving the equation $f(x) = 0$

(a) form an equation, with integer coefficients, that has roots $\frac{2\alpha}{\beta}$ and $\frac{2\beta}{\alpha}$ **(6)**

$g(x) = 4x^2 + px + q$ where p and q are constants

Given that the equation $g(x) = 0$ has roots $3\alpha + \beta$ and $\alpha + 3\beta$

(b) find the value of p and the value of q **(5)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

9. Question 6 · part a · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 8 · part c · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

8 (a) Using the binomial expansion, or otherwise, find the complete expansion of

$$(x + y)^3$$

(1)

The quadratic equation

$$2x^2 + 3x + 4 = 0$$

has roots α and β

(b) Without solving the equation, find the value of

$$\alpha^3 + \beta^3$$

(4)

(c) Hence, form a quadratic equation with integer coefficients that has roots

$\frac{\alpha}{\beta^2}$ and $\frac{\beta}{\alpha^2}$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 8 · part c · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 8 · part c · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 8 · part c · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

(Total for Question 8 is 10 marks)

10. Question 8 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 2 · part whole · 8 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

2 The quadratic equation $3x^2 - 5x + 1 = 0$ has roots α and β

Without solving the equation,

form a quadratic equation with integer coefficients, that has roots $\frac{\alpha}{2\beta}$ and $\frac{\beta}{2\alpha}$ **(8)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

11. Question 2 · part whole · 8 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 2 is 8 marks)

11. Question 2 · part whole · 8 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 5 · part d · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

5 (a) Show that $(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = \alpha^3 + \beta^3$

(2)

The roots of the equation $2x^2 + 3x + 6 = 0$ are α and β

Without solving the equation,

(b) find the value of $\alpha^3 + \beta^3$

(2)

(c) Show that $(\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2 = \alpha^4 + \beta^4$

(2)

(d) Form a quadratic equation with integer coefficients that has roots $(\alpha^3 - \beta)$ and $(\beta^3 - \alpha)$

(6)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

12. Question 5 · part d · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 6 0 4 7 6 A 0 1 3 4 0

13

Turn over ▶

12. Question 5 · part d · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

12. Question 5 · part d · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 5 is 12 marks)

12. Question 5 · part d · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 1 · part whole · 7 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

1 The roots of the equation $4x^2 - 3x - 8 = 0$ are α and β

Without solving this equation, form a quadratic equation, with integer coefficients, which has roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

(Total for Question 1 is 7 marks)

13. Question 1 · part whole · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 6 · part b · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

6 $f(x) = 2x^2 + 5x - p$

The equation $f(x) = 0$ has roots α and β

Given that $\alpha^3 + \beta^3 = -\frac{215}{8}$

(a) find the value of p

(5)

Without solving the equation $f(x) = 0$

(b) form a quadratic equation, with integer coefficients, that has roots

$$\frac{\alpha + \beta}{\alpha^2} \text{ and } \frac{\alpha + \beta}{\beta^2}$$

(5)

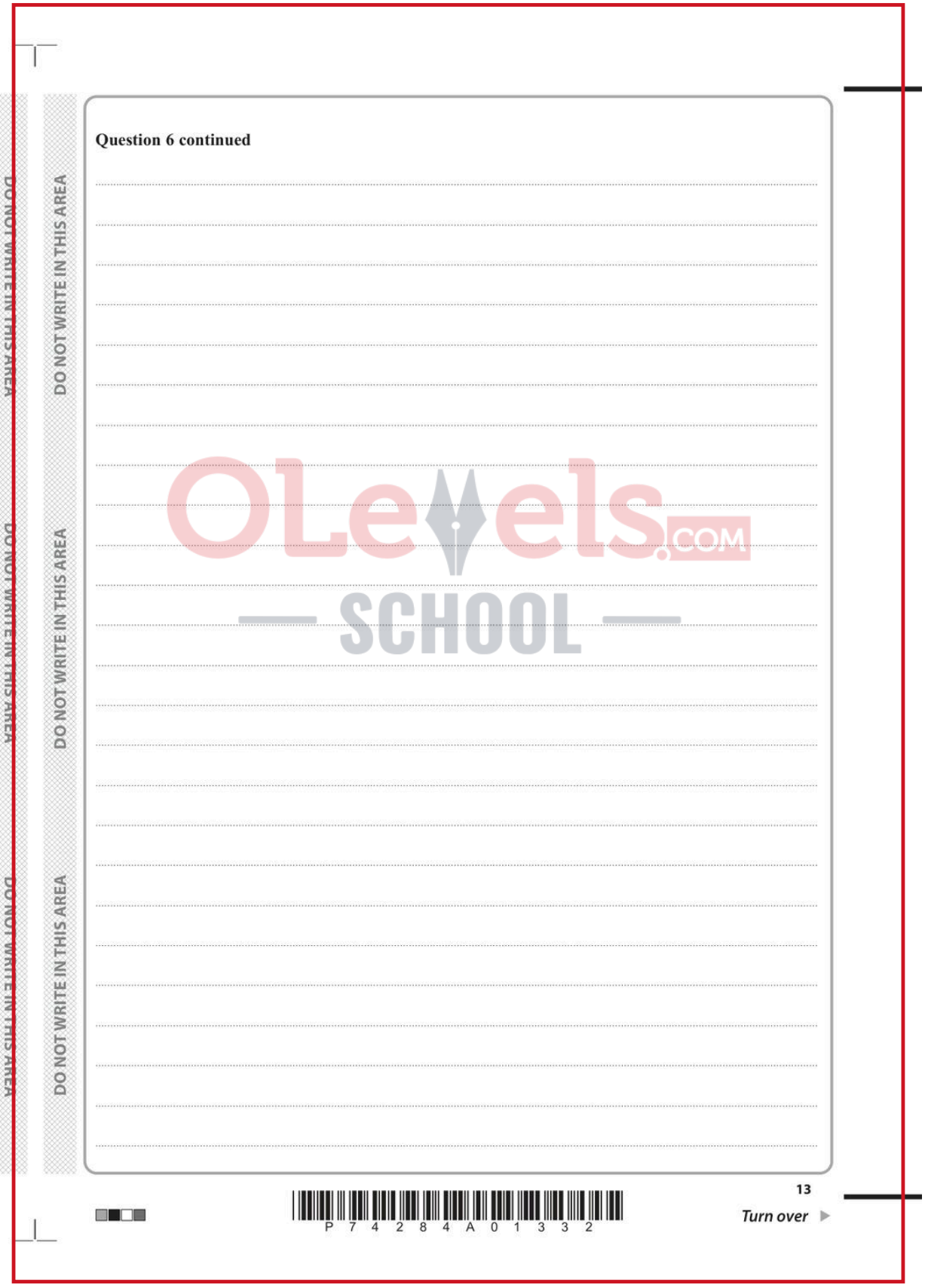
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 4 2 8 4 A 0 1 2 3 2

14. Question 6 · part b · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 4 2 8 4 A 0 1 3 3 2

13

Turn over

14. Question 6 · part b · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 6 continued

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

14. Question 6 · part b · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 6 continued

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 6 is 10 marks)

14. Question 6 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

15. Question 10 · part c · 6 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

10 $f(x) = 3 - 4x - 9x^2$

Given that $f(x)$ can be expressed in the form $A - B(x + C)^2$ where A, B and C are positive constants

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence write down the maximum value of $f(x)$

(1)

The equation $f(x) = 0$ has roots α and β

Without solving the equation $f(x) = 0$

(c) form a quadratic equation, with integer coefficients, that has roots $\frac{3\alpha}{\beta}$ and $\frac{3\beta}{\alpha}$

(6)

(d) Show that $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$

(1)

$g(x) = 3x^2 + qx + r$ where q and r are constants

The equation $g(x) = 0$ has roots $\alpha^2 - \beta$ and $\beta^2 - \alpha$ where α and β are the roots of the equation $f(x) = 0$

(e) Using your answer to part (d), find in simplified exact form, the value of q and the value of r

(6)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

15. Question 10 · part c · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

16. Question 5 · part a · 2 marks

2020 JAN · Paper 02

Further practice · Attempt part a only.

5 Given that α and β are such that $\alpha + \beta = \frac{7}{2}$ and $\alpha\beta = 2$

(a) form a quadratic equation with integer coefficients that has roots α and β,

(2)

(b) form a quadratic equation with integer coefficients that has roots $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.

(6)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

16. Question 5 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

17. Question 5 · part b · 6 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

5 Given that α and β are such that $\alpha + \beta = \frac{7}{2}$ and $\alpha\beta = 2$

(a) form a quadratic equation with integer coefficients that has roots α and β,

(2)

(b) form a quadratic equation with integer coefficients that has roots $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.

(6)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

17. Question 5 · part b · 6 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 5 is 8 marks)

17. Question 5 · part b · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

18. Question 8 · part c · 8 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

8 $f(x) = 2x^2 - 4x + 9$

The curve C has equation $y = f(x)$
The line l with equation $y - 8x + 9 = 0$ is a tangent to C
The line k is perpendicular to l and is also the tangent to C at the point where $x = p$

(a) Find the value of p

(5)

(b) Show that

(i) $(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$

(ii) $\alpha^4 + \beta^4 = \left((\alpha + \beta)^2 - 2\alpha\beta\right)^2 - 2(\alpha\beta)^2$

(2)

The quadratic equation $f(x) = 0$ has roots α and β
Without solving the equation and using your results from part (b)

(c) form a quadratic equation with integer coefficients, that has roots

$\alpha^3 - \beta$ and $\beta^3 - \alpha$

(8)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 6 5 0 7 A 0 1 8 3 2

18. Question 8 · part c · 8 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

18. Question 8 · part c · 8 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

18. Question 8 · part c · 8 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 8 is 15 marks)

18. Question 8 · part c · 8 marks · Working and answer

Working and answer

OLevels.com SCHOOL

19. Question 6 · part b · 5 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

6 $f(x) = 4x^2 - 3x - 5$

The equation $f(x) = 0$ has roots α and β

Without solving the equation $f(x) = 0$

(a) form an equation, with integer coefficients, that has roots $\frac{2\alpha}{\beta}$ and $\frac{2\beta}{\alpha}$ **(6)**

$g(x) = 4x^2 + px + q$ where p and q are constants

Given that the equation $g(x) = 0$ has roots $3\alpha + \beta$ and $\alpha + 3\beta$

(b) find the value of p and the value of q **(5)**

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 6 2 2 8 5 A 0 1 2 3 2

19. Question 6 · part b · 5 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

19. Question 6 · part b · 5 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

19. Question 6 · part b · 5 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

(Total for Question 6 is 11 marks)

19. Question 6 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

20. Question 8 · part a · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

8

$$f(x) = 3x^2 - x + 4$$

$$g(x) = x^2 - px + q$$

The roots of the quadratic equation $f(x) = 0$ are α and β

The roots of the quadratic equation $g(x) = 0$ are $\left(\alpha + \frac{1}{\alpha}\right)$ and $\left(\beta + \frac{1}{\beta}\right)$

Without solving the equation $f(x) = 0$

(a) show that $p = \frac{7}{12}$

(3)

(b) Find the value of q

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

20. Question 8 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

21. Question 8 · part b · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

8

$$f(x) = 3x^2 - x + 4$$

$$g(x) = x^2 - px + q$$

The roots of the quadratic equation $f(x) = 0$ are α and β

The roots of the quadratic equation $g(x) = 0$ are $\left(\alpha + \frac{1}{\alpha}\right)$ and $\left(\beta + \frac{1}{\beta}\right)$

Without solving the equation $f(x) = 0$

(a) show that $p = \frac{7}{12}$

(3)

(b) Find the value of q

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

21. Question 8 · part b · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

21. Question 8 · part b · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

21. Question 8 · part b · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 8 is 7 marks)

21. Question 8 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

22. Question 10 · part e · 6 marks

2024 NOV · Paper 01

Further practice · Attempt part e only.

10 $f(x) = 3 - 4x - 9x^2$

Given that $f(x)$ can be expressed in the form $A - B(x + C)^2$ where A, B and C are positive constants

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence write down the maximum value of $f(x)$

(1)

The equation $f(x) = 0$ has roots α and β

Without solving the equation $f(x) = 0$

(c) form a quadratic equation, with integer coefficients, that has roots $\frac{3\alpha}{\beta}$ and $\frac{3\beta}{\alpha}$

(6)

(d) Show that $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$

(1)

$g(x) = 3x^2 + qx + r$ where q and r are constants

The equation $g(x) = 0$ has roots $\alpha^2 - \beta$ and $\beta^2 - \alpha$ where α and β are the roots of the equation $f(x) = 0$

(e) Using your answer to part (d), find in simplified exact form, the value of q and the value of r

(6)

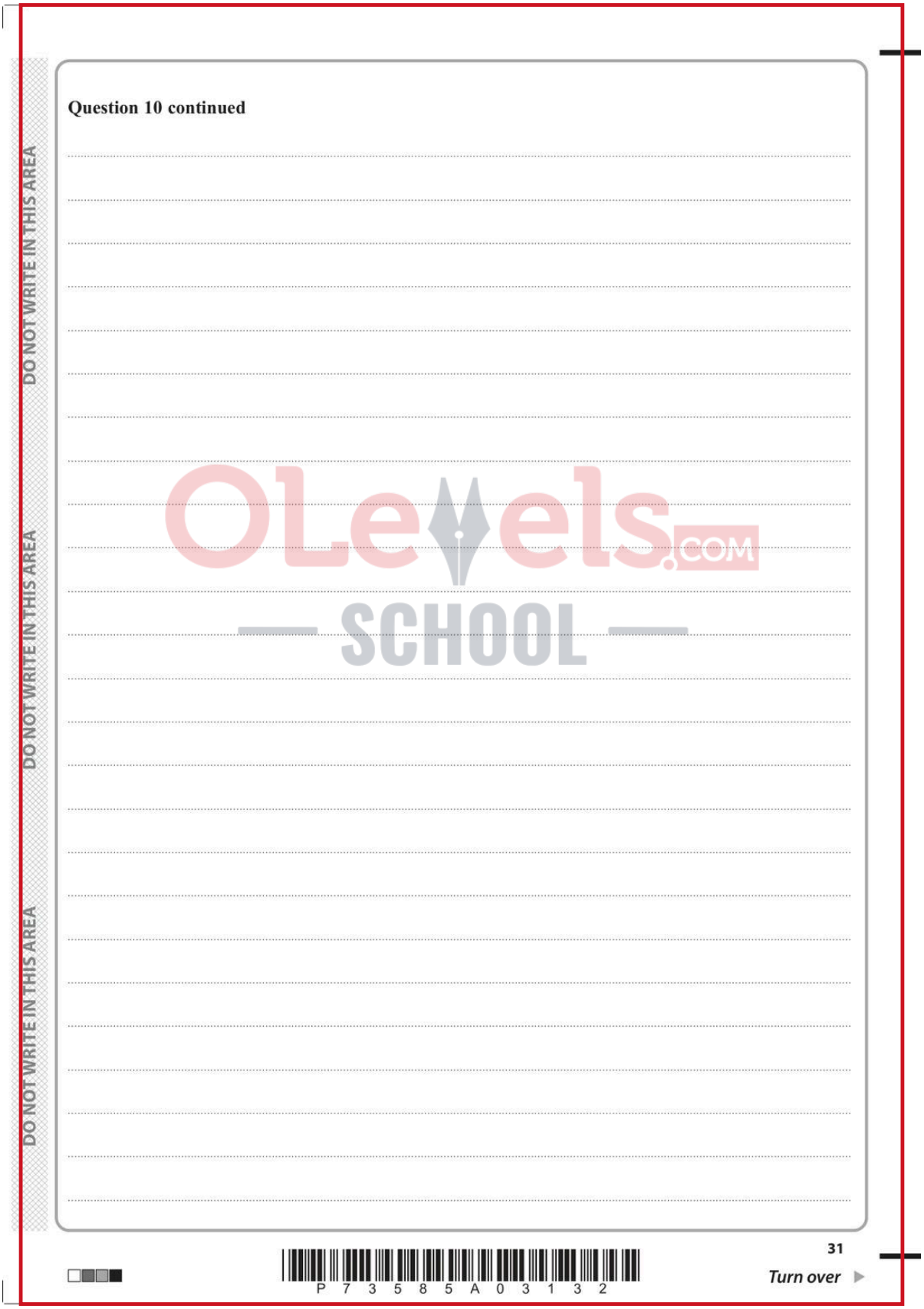
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

22. Question 10 · part e · 6 marks

2024 NOV · Paper 01

Further practice · Attempt part e only.



Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 3 5 8 5 A 0 3 1 3 2

31

Turn over ▶

22. Question 10 · part e · 6 marks

2024 NOV · Paper 01

Further practice · Attempt part e only.

Question 10 continued

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 10 is 18 marks)

22. Question 10 · part e · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL